

Privacy of Numeric Queries Via Simple Value Perturbation

The Laplace Mechanism

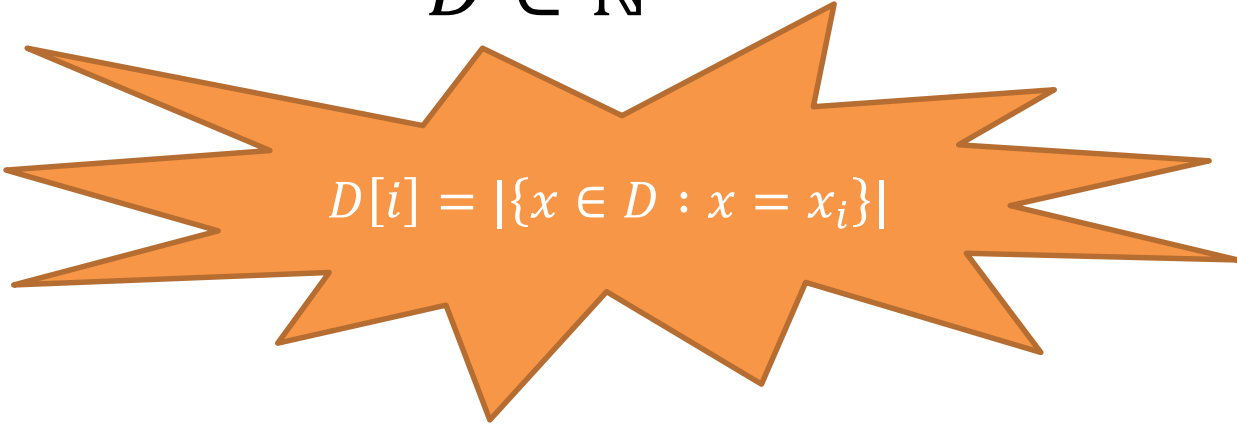
Slides are extracted from material by Aaron Roth

Differential Privacy

A Basic Model

- Let X represent an abstract data universe and D be a multi-set of elements from X .
 - i.e. D can contain multiple copies of an element $x \in X$.
- Convenient to represent D as a *histogram*:

$$D \in \mathbb{N}^{|X|}$$


$$D[i] = |\{x \in D : x = x_i\}|$$

Differential Privacy

A Basic Model

- i.e for a database of heights
 - $D = \{5'2, 6'1, 5'8, 5'8, 6'0\} \subset [4 - 8]$
 - $D = (\dots, \underbrace{1, 0, 0, 0, 0, 0}_{5'2}, \underbrace{2, 0, 0, 0}_{5'8}, \underbrace{1, 1, 0}_{6'0 \ 6'1}, \dots) \in \mathbb{R}^{48}$

Differential Privacy

A Basic Model

- The *size* of a database n :
 - As a set: $n = |D|$.
 - As a histogram: $n = ||D||_1 = \sum_{i=1}^{|X|} |D[i]|$

Definition: ℓ_1 (Manhattan) Distance.

For $\hat{v} \in \mathbb{R}^d$, $||\hat{v}||_1 = \sum_{i=1}^d |\hat{v}_i|$.

Differential Privacy

A Basic Model

- The *distance* between two databases:
 - As a set: $|D \Delta D'|$.
 - As a histogram: $\|D - D'\|_1$

Differential Privacy

A Basic Model

- i.e for a database of heights
 - $- D = \{5'2, 6'1, 5'8, 5'8, 6'0\} \subset [4 - 8]$
 - $- D = (\dots, \underbrace{1,0,0,0,0,0}_{5'2}, \underbrace{2,0,0,0}_{5'8}, \underbrace{1,1,0}_{6'0 \ 6'1}, \dots) \in \mathbb{R}^{48}$
 - $- D' = (\dots, 2,1,0,0,0,0,1,0,0,0,1,1,0, \dots) \in \mathbb{R}^{48}$

$$||D||_1 = |1| + |2| + |1| + |1| = 5$$

$$||D'||_1 = |2| + |1| + |1| + |1| + |1| = 6$$

$$||D - D'||_1 = |-1| + |-1| + |1| = 3$$

Basic Lower Bound: Blatant Non-Privacy

- How much noise is necessary to guarantee privacy?
- A simple model.
 - For simplicity, $D \in \{0,1\}^{|X|}$ (i.e. no repeated elts)
 - A query is a bit vector $Q \in \{0,1\}^{|X|}$
 - $Q(D) = \langle Q, D \rangle = \sum_{i:Q[i]=1} D[i]$
 - A “subset sum query”
 - For $S \subseteq [n]$ write Q_S for the vector:
$$Q_S[i] = \begin{cases} 1, & i \in S \\ 0, & i \notin S \end{cases}$$

Differential Privacy

A Basic Model

Definition: A randomized algorithm with domain $\mathbb{N}^{|X|}$ and range R

$$M: \mathbb{N}^{|X|} \rightarrow R$$

is (ϵ, δ) -*differentially private* if:

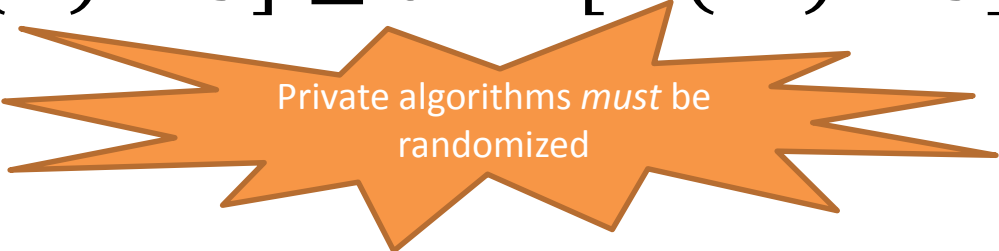
1) For all pairs of databases $D, D' \in \mathbb{N}^{|X|}$ such that $\|D - D'\|_1 \leq 1$ and,



Differing in 1 person's data

2) For all *events* $S \subseteq R$:

$$\Pr[M(D) \in S] \leq e^\epsilon \Pr[M(D') \in S] + \delta.$$



Private algorithms *must* be randomized

Resilience to Post Processing

Proposition: Let $M: \mathbb{N}^{|X|} \rightarrow R$ be (ϵ, δ) -differentially private and let $f: R \rightarrow R'$ be an arbitrary function. Then:

$$f \circ M: \mathbb{N}^{|X|} \rightarrow R'$$

is (ϵ, δ) -differentially private.



Thinking about the output of M can't make it less private.

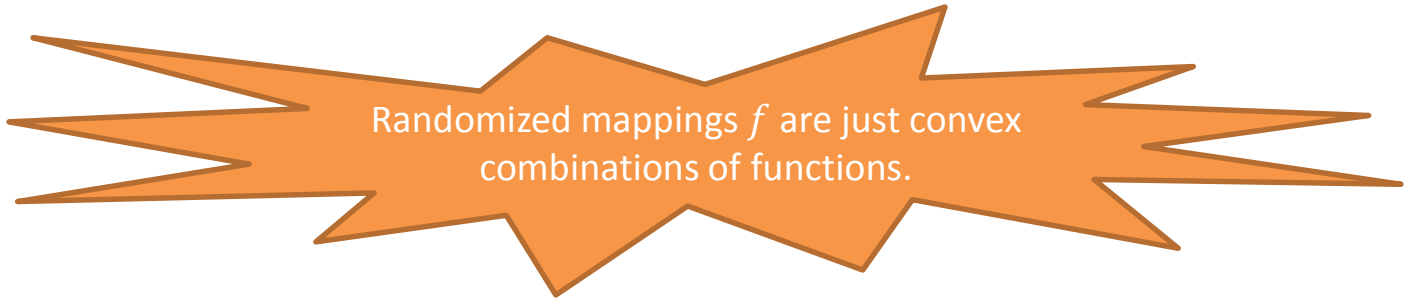
Resilience to Post Processing

Proof:

- 1) Consider any pair of databases $D, D' \in \mathbb{N}^{|X|}$ with $\|D - D'\|_1 \leq 1$.
- 2) Consider any event $S \subseteq R$.
- 3) Let $T \subseteq R$ be defined as $T = \{r \in R : f(r) \in S\}$.

Now:

$$\begin{aligned}\Pr[f(M(D)) \in S] &= \Pr[M(D) \in T] \\ &\leq e^\epsilon \Pr[M(D') \in T] + \delta \\ &= e^\epsilon \Pr[f(M(D')) \in S] + \delta\end{aligned}$$



Randomized mappings f are just convex combinations of functions.

Resilience to Post Processing

Take away message:

- 1) f as the adversaries analysis: can incorporate arbitrary auxiliary information the adversary may have. Privacy guarantee holds no matter what he does.
- 2) f as our algorithm: If we access the database in a differentially private way, we don't have to worry about how our algorithm post-processes the result. *We only have to worry about the data access steps.*

Answering Numeric Queries

- Suppose we have some numeric *question* about the private database that we want to know the answer to:

$$Q: \mathbb{N}^{|X|} \rightarrow \mathbb{R}^k. \quad Q(D) = ?$$

- How do we do it privately?
 - How much noise do we have to add?
 - What are the relevant properties of Q ?

Answering Numeric Queries

Definition: The ℓ_1 -sensitivity of a query

$Q: \mathbb{N}^{|X|} \rightarrow \mathbb{R}^k$ is:

$$GS(Q) = \max_{D, D': ||D - D'||_1 \leq 1} ||Q(D) - Q(D')||_1$$

i.e. how much can 1 person affect the value of the query?

“How many people in this room have brown eyes”: Sensitivity 1

“How many have brown eyes, how many have blue eyes, how many have green eyes, and how many have red eyes”: Sensitivity 1

“How many have brown eyes and how many are taller than 6”: Sensitivity 2

Answering Numeric Queries

The Laplace Distribution:

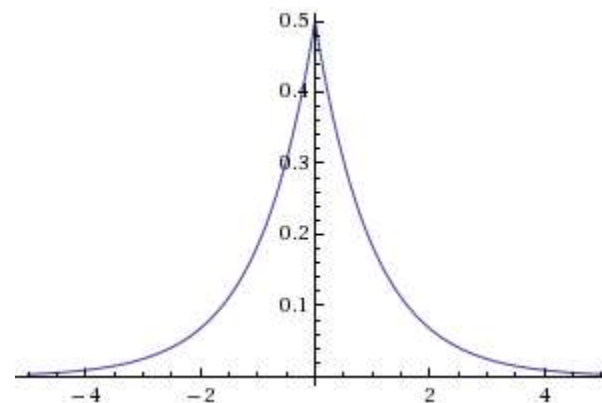
$\text{Lap}(b)$ is the probability distribution with p.d.f.:

$$p(x | b) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

i.e. a symmetric exponential distribution

$$Y \sim \text{Lap}(b), \quad E[|Y|] = b$$

$$\Pr[|Y| \geq t \cdot b] = e^{-t}$$



Answering Numeric Queries: The Laplace Mechanism

$\text{Laplace}(D, Q: \mathbb{N}^{|X|} \rightarrow \mathbb{R}^k, \epsilon):$

1. Let $\Delta = GS(Q)$.
2. For $i = 1$ to k : Let $Y_i \sim \text{Lap}(\frac{\Delta}{\epsilon})$.
3. Output $Q(D) + (Y_1, \dots, Y_k)$

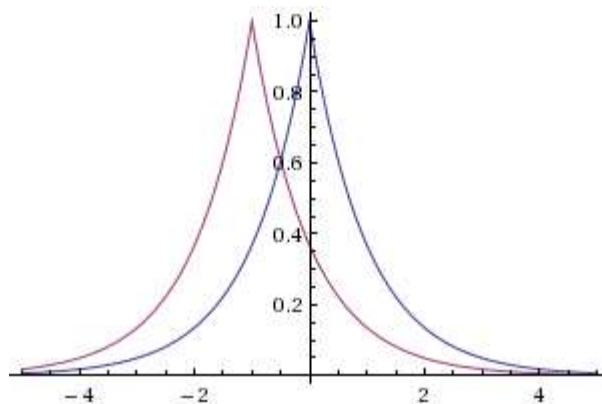
Independently perturb each coordinate of the output with Laplace noise scaled to the sensitivity of the function.

Idea: This should be enough noise to hide the contribution of any single individual, no matter what the database was.

Answering Numeric Queries: The Laplace Mechanism

$\text{Laplace}(D, Q: \mathbb{N}^{|X|} \rightarrow \mathbb{R}^k, \epsilon)$:

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To Ponder

- Where is there room for improvement?
 - The Laplace mechanism adds *independent* noise to every coordinate...
 - What happens if the user asks (essentially) the same question in every coordinate?
 - Read [Dinur,Nissim03]: a computationally efficient attack that gives blatant non-privacy for a mechanism that adds noise bounded by $o(\sqrt{n})$.